

# Capstone Impact Report

## 1. Purpose

This report measures the impact of right-sizing the agentic engineering pipeline and records the evidence available before and during the final human-approved production-like capstone run.

The report separates measured results from metrics that still require additional runtime evidence. No unsupported metric is estimated or invented.

## 2. Measured Deterministic Conversion

One stable workflow step, handoff validation, was converted from agentic execution to deterministic code.

| Metric                       | Before: Agentic Validation | After: Deterministic Validation | Result            |
|------------------------------|----------------------------|---------------------------------|-------------------|
| Average cycle time           | 45 seconds                 | 0.2 seconds                     | Improved          |
| Token cost per run           | \$0.003                    | \$0                             | Improved          |
| Review latency               | ~30 seconds                | ~5 seconds                      | Improved          |
| Deterministic harness checks | 7/7 passing                | 7/7 passing                     | Quality preserved |
| Output variance              | Not recorded               | Zero across 3 runs              | Repeatable        |

## 3. Required Five-Metric Before/After Summary

The following summary reports only supported values. Where a trustworthy baseline is unavailable, the limitation is stated rather than replaced with an estimate.

| Required Metric | Before                                               | After                                                                                                                         | Evidence / Interpretation                                                                 |
|-----------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Quality         | 7/7 deterministic checks passing                     | 7/7 deterministic checks passing; later integrated regression suite 24/24 passing                                             | Measured quality was preserved while the validation step was simplified.                  |
| Review latency  | ~30 seconds                                          | ~5 seconds                                                                                                                    | Measured for the handoff-validation path.                                                 |
| Defect rate     | No reliable baseline defect-rate percentage recorded | No general defect-rate percentage claimed; successful run recorded 0 rejected evaluation items across 8 recorded review items | Evidence supports a successful evaluated run, not a general 0% defect-rate claim.         |
| Cycle time      | 45 seconds                                           | 0.2 seconds                                                                                                                   | Measured for handoff validation after conversion from agentic to deterministic execution. |
| Cost            | ~\$0.003 model-token cost per validation run         | \$0 model-token cost for the deterministic validation step                                                                    | Applies only to the converted validation step, not the complete multi-agent workflow.     |

## 4. Governance Impact

Calibration identified governance near-misses including overly broad Implementer delete access, Reviewer access that could trigger an inappropriate testing capability, and retrieval above the Implementer's classification ceiling.

The final design responds through role-specific MCP allow-lists, data-classification ceilings, read-only role boundaries, policy tests, governed-file tests, red-team testing, and human checkpoints.

## 5. Final Human-Approved Production-Like Run Evidence

| Metric                | Result                                                                 |
|-----------------------|------------------------------------------------------------------------|
| Model                 | anthropic/claude-haiku-4.5                                             |
| Agent path            | Planner → Implementer → Reviewer → Tester                              |
| Full-pipeline runtime | 46.8 seconds                                                           |
| Workflow token budget | 33,519 / 38,000 tokens                                                 |
| Retrieval             | Deterministic lexical matching with classification-ceiling enforcement |
| Human approval        | approve                                                                |

| Metric                   | Result                                      |
|--------------------------|---------------------------------------------|
| Completion authorization | true                                        |
| Transcript               | logs/capstone-final-human-approved-run.json |
| Audit evidence           | logs/capstone-final-run-002.log             |

The successful run exercised deterministic lexical retrieval, persistent-storage reads, an Implementer storage write, Reviewer verification, and Tester validation. Recorded evaluation items were approved; this is evidence of a successful evaluated run, not a general defect-rate claim.

## 6. Tool-Evolution Drill

A controlled permission-change drill temporarily revoked the documentation-writer role's granted read\_entry permission. The policy suite changed from 6 passing tests to 1 failing and 5 passing; after the permission was restored, all 6 tests passed again.

This demonstrated that the evaluation and governance system can detect permission drift between documented policy and executable MCP enforcement.

## 7. Limitations and Interpretation

The final transcript does not establish a trustworthy full-pipeline model-cost measurement, so no unsupported end-to-end cost claim is made. One successful evaluated run is not sufficient evidence to claim a general 0% defect rate across future workloads.

Human-checkpoint conditions are defined in governance policy. The successful final run recorded an explicit human approval decision of approve and completion\_authorized=true; it did not require an escalation condition.

## 8. Conclusion

The available evidence shows that deterministic conversion reduced latency and model-token cost for a stable workflow step without detected quality regression. The final governed workflow also demonstrates scoped MCP permissions, evaluation gates, audit evidence, and an explicit human approval checkpoint.